

Chapter 102 — Surgical, Radiographic, and Endoscopic Anatomy of the Male Pelvis

BONY PELVIS

- **Composition:** the sacrum posteriorly and the paired **innominate (hip) bones** anteriorly and laterally.
- Each innominate bone is **three fused bones** — **ilium, ischium and pubis** — fused at the acetabulum: ilium → upper acetabulum; ischium → posteroinferior acetabulum; pubis → anterior acetabulum.
- **Joints:** sacroiliac (SI) joints posteriorly; a **cartilaginous joint at the pubic symphysis** anteriorly.
- Two strong ligaments fix the sacrum to the hip bones, allowing only limited upward movement of the sacrum and so giving the SI joint resilience when the vertebral column takes sudden increases in weight. They also create the sciatic foramina:

SACROTUBEROUS VERSUS SACROSPINOUS LIGAMENT

	Sacrotuberous	Sacrospinous
Position	More posterior	More anterior
Attachments	Sacrum/coccyx → ischial tuberosity	Sacrum/coccyx → ischial spine
Foramina	Forms the common sciatic foramen	Bisects it into greater and lesser sciatic foramina

- **Orientation:** standing, the pelvis curves obliquely backward relative to the trunk and is tilted forward; the **plane of the pelvic brim lies at 50–60° to the horizontal**.
- **Greater (false) pelvis** — the iliac fossae, extending up to the iliac crest; **lesser (true) pelvis** — contains the urogenital organs. They are separated by the **arcuate line**, running from the sacral promontory to the pubis, where it becomes continuous with the pectineal line.
- **Pelvic outlet (inferior pelvic aperture):** an incomplete, roughly circular bony line — lower symphysis pubis, ischiopubic rami, ischial tuberosities, sacrotuberous ligaments and coccyx.
- **Bony landmarks at urological surgery:** pubic arch and symphysis; ischial tuberosities; the **iliac crest at the level of L4**, palpable as it runs posteriorly from the anterior superior iliac spine (ASIS).
- **Cooper's (pectineal) ligament** overlies the pectineal line of the pubis and takes sutures in inguinal hernia repair and urethral suspension procedures.
- **Pelvic radiographs and fluoroscopy** show the bony landmarks in high resolution and detail. *Figure 102.2* — an anteroposterior (AP) radiograph (iliac wing, sacrum, sciatic notch, iliopectineal line, obturator foramen, ischial tuberosity, a phlebolith) and an oblique voiding cystourethrography spot image showing the bladder, verumontanum, prostatic urethra, membranous urethra and an anterior urethral stricture.

Concept — *The pelvis is three fused bones braced by two ligaments: the ligaments stabilise the SI joint and, in doing so, create the sciatic foramina through which pelvic structures leave; the arcuate line then divides the abdominal (false) pelvis from the surgical (true) pelvis.*

LOWER ABDOMINAL WALL

Skin and Subcutaneous Fascia

- The **umbilicus marks the T10 dermatome** and usually lies at the **L3 vertebra**; its position varies with the amount of subcutaneous fat.

- **Semilunar lines** (visible with good muscle definition) mark the lateral borders of the rectus abdominis and its sheath, running from the inferior costal margin near the ninth costal cartilages to the pubic tubercles. Three transverse skin grooves may overlie the rectus tendinous intersections.
- The **inguinal groove** marks the inguinal ligament; the groove just inferior and parallel to it marks the division between the anterior abdominal wall and the thigh.

CAMPER'S VERSUS SCARPA'S FASCIA

	Camper's fascia	Scarpa's fascia
Layer	Superficial fatty	Deep membranous ; blends superficially and laterally with Camper's
Contents / continuity	Superficial circumflex iliac, external pudendal and superficial inferior epigastric vessels (branches of the femoral vessels)	Inferiorly fuses with fascia lata ~1 cm below and parallel to the inguinal ligament ; medially continuous with Colles fascia
Surgical relevance	Superficial inferior epigastric vessels met at inguinal incisions and laparoscopic trocar placement	Defines the compartment for extravasation and infection (below)

- **Colles fascia** fuses laterally to the **ischiopubic rami** and posteriorly to the **posterior edge of the perineal membrane**, and is continuous with the **dartos fascia** of penis and scrotum (Figure 102.3).
- These fusions **limit the spread of necrotising soft tissue infection of the scrotum**, and of urinary extravasation and bleeding after **anterior urethral injury**.
- After anterior urethral injury blood collects in scrotum and penis **deep to dartos**, confined laterally by the Colles–ischiopubic fusion → the **"butterfly" haematoma**.
- The fasciae stop spread **down the leg or into the buttocks** but **not superiorly**: in the space deep to Scarpa's fascia, spread continues **up the abdominal wall, around the flank to the back**.
- **Figure 102.3** — left, the deep fasciae of abdomen, perineum and thigh (Scarpa's, Colles, fascia lata), with the superficial inferior epigastric vessels running superiorly in Camper's fascia; right, a midline sagittal view of the pelvic fasciae — Denonvilliers fascia, striated urethral sphincter, perineal membrane, Buck's fascia, dartos and central tendon.

Musculature

- The lower abdominal wall muscles lie **deep to Scarpa's fascia**. External oblique, internal oblique and transversus abdominis end medially as aponeuroses that **fuse in the midline at the linea alba** (landmark for midline incisions) and form the rectus sheath.

RECTUS SHEATH ABOVE VERSUS BELOW THE ARCUATE LINE

	Above the arcuate line	Below the arcuate line
Anterior sheath	External oblique aponeurosis + part of internal oblique aponeurosis	All three aponeuroses pass anterior to rectus
Posterior sheath	Remaining internal oblique aponeurosis + transversus aponeurosis	Absent — rectus covered posteriorly only by transversalis fascia and peritoneum
Internal oblique aponeurosis	Splits around rectus	Does not split; fuses anteriorly with external oblique and transversus aponeuroses

- The **arcuate line** lies about **one-third of the distance from the umbilicus to the pubic crest**. Above it, the anterior and posterior layers unite medially to form the linea alba (Figure 102.4).
- **Rectus abdominis**: pubis (medial to the pubic tubercle) → xiphoid and adjacent costal cartilages. Fibres run between **three or more tendinous intersections** — at the xiphoid, the umbilicus and halfway between

- each **firmly attached to the anterior rectus sheath**, so the muscle **can be divided transversely without significant retraction**.
- **Nerve supply:** thoracic segmental nerves entering **laterally**; dividing them in a **paramedian incision lateral to the rectus** can cause rectus atrophy and predispose to **ventral hernia**.
- **Pyramidalis:** small triangular muscle, **absent in about 20% of people**, anterior to rectus but within its sheath; pubic crest → linea alba, which it tenses (Figure 102.5A).

Concept — Fascial fusions decide where fluid goes: *the Scarpa-Colles-dartos continuum is sealed to the thigh, the ischiopubic rami and the perineal membrane but open superiorly. The rectus is segmented by intersections tethered to the anterior sheath yet innervated from the side — so transverse division is safe and lateral paramedian division is not.*

Inguinal Canal

- In adults, an **inferomedially directed oblique passage ~4 cm long** through the lower anterior abdominal wall, transmitting the **spermatic cord and ilioinguinal nerve**.

DEEP VERSUS SUPERFICIAL INGUINAL RING

	Deep (internal) ring	Superficial (external) ring
Nature	Evagination of transversalis fascia ; entrance to canal	Slit-like opening in external oblique aponeurosis ; exit from canal
Position	Above inguinal ligament, lateral to inferior epigastric vessels ; midway between ASIS and pubic tubercle ; 4 cm lateral to external ring	Superolateral to pubic tubercle

BOUNDARIES OF THE INGUINAL CANAL

Wall	Formed by
Anterior	External oblique aponeurosis throughout; reinforced laterally by the lowest internal oblique fibres
Posterior	Transversalis fascia ; posteromedially reinforced at the external ring by the conjoint tendon (inguinal falx) — merged internal oblique and transversus aponeuroses (Figure 102.6)
Roof	Internal oblique and transversus fibres arching over the canal to fuse at the conjoint tendon
Floor	Laterally iliopubic tract ; centrally the gutter-like superior surface of the inguinal ligament ; medially the lacunar ligament

- **Shutter mechanism:** contraction of internal oblique and transversus **closes the roof against the floor**, preventing herniation of intra-abdominal contents.
- **Direct hernia — medial to the inferior epigastric vessels; indirect hernia — lateral to them.** *Figure 102.5B–C* — the canal viewed from the preperitoneal space, showing indirect, direct and femoral hernia sites relative to the inferior epigastric vessels, iliopubic tract, Cooper's ligament and external iliac vessels.
- **Radical orchiectomy:** the canal is opened and the cord **ligated at the internal ring**, allowing the proximal cord to retract into the retroperitoneum.
- *Figure 102.6* — coverings of the cord in cross section: external spermatic fascia, cremaster (within cremasteric fascia) and internal spermatic fascia.

Internal Peritoneal Elevations

- The inner lower abdominal wall is covered by transversalis fascia, a variable amount of extraperitoneal fat and parietal peritoneum. Below the umbilicus it shows **peritoneal folds, some containing fetal vessel remnants**, easily identified at transperitoneal laparoscopic and robotic surgery (Figure 102.7).

UMBILICAL FOLDS OF THE LOWER ABDOMINAL WALL (MEDIAL TO LATERAL)

Fold	Covers	Remnant / structure	Surgical significance
Median (single)	Median umbilical ligament	Urachus — fetal bladder apex → umbilicus	Runs from bladder apex to umbilicus
Medial (paired)	Medial umbilical ligaments	Obliterated umbilical arteries , traceable to the internal iliac artery	Incise peritoneum lateral to them → obturator fossa and retropubic space
Lateral (paired)	Inferior epigastric vessels	Live vessels ascending to supply rectus	Lateral boundary of the medial fold territory

KEY POINTS

- The bony pelvis consists of three fused bones: ilium, ischium and pubis.
- The true and false pelvis are separated by the arcuate line.
- The inguinal canal transmits the spermatic cord and the ilioinguinal nerve.
- The medial and median umbilical ligaments are remnant fetal structures and landmarks for access to the retropubic space at transperitoneal laparoscopic and robotic surgery.

Concept — *The inferior epigastric vessels are the organising landmark of the lower wall: they mark the deep ring, divide direct from indirect hernia, and form the lateral fold that completes the laparoscopic map — urachus, umbilical arteries, epigastrics, from the midline outwards.*

SOFT TISSUES OF THE PELVIS**Musculature**

- The lateral pelvic sidewalls are padded by **obturator internus**, which leaves via the **lesser sciatic foramen** to insert on the femur. Its medial surface carries obturator fascia, **thickened as a tendinous arch from the ischial spine to the lower half of the pubis**.
- This **tendinous arch of the levator ani** is the origin of the bowl-shaped **pelvic diaphragm = levator ani + coccygeus** (Figure 102.8).

MUSCLES OF THE PELVIC DIAPHRAGM

Muscle	Position	Attachments / form	Function
Coccygeus	Most posterior; smallest	Ischial spine → inferior sacrum and coccyx	Supports viscera; flexes coccyx
Puborectalis (puboanal sling in Figure 102.8)	Most anterior part of levator ani	U-shaped sling passing posterior to the anorectal junction	Major role in faecal continence
Pubococcygeus	Intermediate	Pubis → coccyx	Floor support
Iliococcygeus	Posterolateral	Tendinous arch → coccyx	Floor support

- Levator ani is under **somatic control**: a dynamic floor, **tonically contracted** to support the viscera and maintain urinary and faecal continence; it contains both **type I (slow-twitch) and type II (fast-twitch) fibres**.
- It **relaxes** to allow defecation and micturition, then **contracts to raise the floor**; further contraction with **Valsalva** resists raised intra-abdominal pressure that would otherwise drive contents through the pelvic outlet.
- **Kegel exercises** to regain or maintain continence after pelvic surgery aim to **increase the tonic contraction** of these muscles.

Pelvic Fascia

- Connective tissue continuous with the retroperitoneal fascia and with the fascia between the muscular and peritoneal layers of the lower abdominal wall.

PARIETAL VERSUS VISCERAL PELVIC FASCIA

	Parietal pelvic fascia	Visceral pelvic fascia
Covers	Pelvic aspect of muscles of the walls and floor ; membranous, variable thickness	Pelvic organs and their neurovascular supply — forms each organ's adventitia
Continuity / fixation	Continuous superiorly with transversalis fascia ; fixed to the arcuate line, Cooper's ligament, sacrospinous ligament, ischial spine and tendinous arch of levator ani	Lateral attachments of organs to sidewall = endopelvic fascia (Figure 102.9)

- **Arcus tendineus fascia pelvis**: a thick white band of condensed connective tissue on the superomedial parietal fascia over levator ani, running from the **puboprostatic ligament to the ischial spine**. It is the **junction of parietal and visceral fascia** and anchors the visceral condensations that **support the urethra and bladder**.
- **Endopelvic fascia** is the **conduit for neurovascular structures** from sidewall to organs and holds the organs in place; its loose tissue extends to the midline, **separating bladder, prostate and seminal vesicles from the rectum**. All pelvic vascular structures and some pelvic nerves travel in it.

Concept — *The floor is a somatically controlled, tonically active muscle bowl; the fascia is organised as a wall lining (parietal) and organ sheaths (visceral) joined at the arcus tendineus — and because the visceral layer is the neurovascular highway, entering the endopelvic fascia is the gateway to the prostate.*

PERINEUM

Perineal Fascia

- **Superficial fatty layer + deep membranous layer (Colles fascia)**. The fatty layer is greatly diminished in males and **replaced in penis and scrotum by smooth (dartos) muscle**; it is continuous between penis/scrotum and thighs with the abdominal subcutaneous fat.
- **Colles fascia attachments**: posteriorly the **posterior margin of the perineal membrane**; laterally the fascia lata of the medial thigh; anteriorly continuous with the dartos of penis and scrotum and with **Scarpa's fascia**.
- **Perineal membrane**: thin, tough, triangular deep fascia enclosing the urogenital triangle; spans the pubic arches (inferior ischiopubic rami) from the **pubis to the ischial tuberosities**; **perforated by the urethra**. Its anterior aspect **supports the urethral sphincter**; the external genitalia attach to its **inferior surface**; superficial and deep transverse perineal muscles run along its posterior edge to the perineal body (Figure 102.10).
- **Perineal body** — **not a structure but an aggregation of fibromuscular tissue** in the midline at the junction of urogenital and anal triangles. Anteriorly it blends with the posterior border of the perineal membrane; superiorly with the **rectovesical septum (contiguous with Denonvilliers fascia)**.
- It is the **focal point of pelvic support** — nearly every pelvic muscle and fascia inserts into it; **injury at perineal prostatectomy risks urinary and possibly faecal incontinence**.
- **Perineum** = both an external surface (perineal region) and a **shallow compartment limited superiorly by the pelvic diaphragm**. Osseofibrous boundaries: **symphysis** anteriorly, **ischiopubic rami** anterolaterally, **ischial tuberosities** laterally, **sacrospinous ligaments** posterolaterally, **coccyx** posteriorly.

- A **transverse line joining the anterior ends of the ischial tuberosities** divides it into the urogenital (anterior) and anal (posterior) triangles.

Anal Triangle

- **Contents:** anal canal and sphincters; ischiorectal fossa with its nerves and vessels. Lined by superficial and deep fascia; perianal subcutaneous fat is continuous with that of the urogenital triangle, buttocks and thigh.
- **Ischiorectal fossa:** levator ani medially; obturator internus and sacrotuberous ligament laterally; fat-filled; extends anteriorly to a recess above the urogenital diaphragm and **posteriorly into the pelvis through the sciatic foramina** → **perirectal infection can spread between perineum and pelvic cavity**.
- The anal canal begins where the rectal ampulla abruptly narrows at the **puborectalis sling**.

INTERNAL VERSUS EXTERNAL ANAL SPHINCTER

	Internal anal sphincter	External anal sphincter
Control	Involuntary smooth muscle	Voluntary , large
Extent	Upper two-thirds of canal	Inferior two-thirds of canal
Origin / parts	Thickening of the inner circular muscle of the rectum	Subcutaneous (to perineal body via the central tendon of the perineum); superficial (perineal body and coccyx); deep (blends with puborectalis)
Innervation	Tone maintained by sympathetic ; lessened by parasympathetic → passive expansion at defecation	Rectal branches of pudendal nerve
Disruption	—	Faecal incontinence

PERINEAL PROSTATECTOMY: YOUNG VERSUS BELT APPROACH

Young procedure	Belt procedure
Divides the central tendon and sphincteric attachments to the perineum, leaving the external sphincter on the rectum	Passes under the external sphincter along the anterior rectal wall

Urogenital Triangle

- Bounded by the **scrotum anteriorly** and the **perineal body posteriorly**. Skin and fat overlie Colles fascia, which attaches posteriorly to the perineal membrane (perineal body at the centre) → the **superficial perineal pouch**.
- **Root of penis** lies deep to this pouch: **corpora cavernosa** attach to the inferior ischiopubic rami and perineal membrane; the **corpus spongiosum** of the bulbar urethra attaches to the perineal membrane.
- **Bulbospongiosus surrounds the bulb; ischiocavernosus surrounds the corpora cavernosa**; their contraction compresses the corpora during erection.
- **Internal pudendal artery** supplies the perineum: from the internal iliac artery it leaves the pelvis through the greater sciatic foramen and enters the perineum through the **lesser sciatic foramen**, running medial to obturator internus in **Alcock's (pudendal) canal** → branches to the anus → through Colles fascia into the superficial pouch and posterior scrotum → **terminates as the dorsal artery of the penis** (Figure 102.11).
- **Venous:** internal pudendal veins anastomose with the **dorsal venous complex** via a plexus on the lateral prostate → **bleeding at prostatic dissection**. Inferior rectal veins (from the internal pudendal vein) anastomose with the middle and superior rectal veins, which drain to the inferior mesenteric vein.

Obstruction of either the portal side (inferior mesenteric vein) or the systemic side (inferior rectal veins) → venous plexus dilatation and haemorrhoids.

- **Pudendal nerve** travels with the internal pudendal vessels in Alcock's canal. The **inferior rectal (rectal) branches** supply the **external anal sphincter and perianal skin**; the **perineal branches** supply the **striated urethral sphincter and levator ani**, the ischiocavernosus and bulbospongiosus, and sensation to the posterior scrotum; the nerve ends anteriorly as the **dorsal nerve of the penis**.

Table 102.1 — normal adult penile length.

TABLE 102.1 PENILE LENGTH IN ADULTS (REBUILT)

Measure	Range of reported means	Notes
Flaccid length	~7.7–10.7 cm	Lowest values in men with erectile dysfunction (ED)
Stretched length	~11.6–16.7 cm	Where both were measured, stretched ≈ erect length
Erect length	~12.9–15.5 cm	Kinsey (1948) : flaccid 9.7 cm, erect 15.5 cm
Populations	United States, France, Italy, Nigeria, Germany, Jordan; ages 17–91 years	
ED versus normal	Men with ED had shorter flaccid and stretched lengths than normal controls	

Modified from Awwad et al., 2005 (penile measurements in normal adult Jordanians and in patients with erectile dysfunction).

- **What the table is for:** normal adult reference ranges. **Pattern:** stretched length is the clinical surrogate for erect length. **Trap:** normal ranges are wide and population-dependent — no single series' mean is a cut-off.

KEY POINTS: PERINEUM

- The perineum is bordered by the symphysis pubis, ischiopubic rami, ischial tuberosities, sacrotuberous ligaments and coccyx.
- It is composed of two triangles: urogenital and anal.
- Its blood supply is the internal pudendal artery and its branches; its innervation is the pudendal nerve.

Concept — The perineum is a **diamond split into two triangles, supplied by one artery and one nerve** (internal pudendal, pudendal) arriving through Alcock's canal; its mechanical **keystone is the perineal body**, which is why perineal approaches are defined by how they handle the central tendon and external sphincter.

PELVIC CIRCULATION

Arterial Supply

- **Middle sacral artery:** from the aortic bifurcation to the anterior sacrum, supplying rectum and sacral foramina; **met in presacral dissection**.
- **Common iliac arteries lie anterior to the common iliac veins** and divide into external and internal iliac arteries at the **level of the SI joint**.
- **External iliac artery:** anterolateral to its vein, on iliopsoas; passes under the inguinal ligament to become the femoral artery. Branches met at **pelvic lymph node dissection (PLND)**: **inferior epigastric** (arises proximal to the inguinal ligament → rectus), **deep circumflex iliac**, and pubic branches. The **cremasteric artery** arises from the inferior epigastric and travels with the cord. **Accessory obturator vessels** branch from the inferior epigastric vessels into the obturator fossa — a PLND hazard.
- **Internal iliac (hypogastric) artery** runs posteriorly to supply the deep pelvis; it divides into **anterior and posterior trunks**. Anterior trunk, in order: **umbilical, obturator, middle rectal and inferior vesical, internal pudendal, then inferior gluteal** (Figure 102.12).

- **Umbilical artery** is obliterated beyond its origin → **medial umbilical ligament**, the landmark for entering the extraperitoneal space. The **superior vesical artery** arises from its proximal (patent) part and supplies the bladder with the inferior vesical. The **artery of the vas deferens** also comes from an umbilical branch and joins the cremasteric and testicular arteries in the cord → an **alternative testicular blood supply**.
- **Obturator artery**: supplies thigh adductors; passes through the obturator foramen; lies **deep and medial to the obturator nerve** — met at PLND (Figure 102.13).

TABLE 102.3 ARTERIES OF THE PELVIS (REBUILT)

Group	Artery	Origin	Supplies
Aortic	Middle sacral	Aorta	Sacral nerves and sacrum
External iliac branches	Inferior epigastric	External iliac	Rectus abdominis and overlying skin and fascia
	Deep circumflex iliac	External iliac	Inguinal ligament and surroundings, laterally
	Pubic	Inferior epigastric	Inguinal ligament and surroundings, medially
Internal iliac — posterior trunk	Cremasteric	Inferior epigastric	Vas deferens and testis
	Superior gluteal	Posterior trunk	Gluteal muscles and overlying skin
	Ascending lumbar	Posterior trunk	Psoas, quadratus lumborum and adjacent structures
	Lateral sacral	Posterior trunk	Sacral nerves and sacrum
Internal iliac — anterior trunk	Superior vesical	Anterior trunk (via proximal umbilical artery)	Bladder, ureter, vas deferens, seminal vesicle
	Middle rectal	Anterior trunk	Rectum, ureter, bladder
	Inferior vesical	Anterior trunk	Bladder, seminal vesicle, prostate, ureter, neurovascular bundle
	Internal pudendal	Anterior trunk	Rectum, perineum, external genitalia
	Obturator	Anterior trunk	Adductors of the thigh and overlying skin
	Inferior gluteal	Anterior trunk	Gluteal muscles and overlying skin

- **What the table is for:** origin-to-territory mapping. **Pattern:** the **posterior trunk is parietal** (gluteal, lumbar, sacral); the **anterior trunk is visceral and perineal**. **Traps:** the deep circumflex iliac arises from the external iliac, not the inferior epigastric; the superior vesical comes off the patent umbilical artery; the inferior vesical, not the internal pudendal, supplies the neurovascular bundle.

Venous Drainage

- A rich venous network. External and internal iliac veins run **inferior and medial to their arteries** and **join behind the internal iliac artery** — injury here at PLND causes **significant bleeding**. An **accessory obturator vein** may drain into the external iliac vein — also met at PLND.
- **Dorsal venous complex (DVC):** drains the penis; enters the pelvis **beneath the pubic symphysis in the midline, anterior to the striated sphincter**; divides into a **central superficial branch** and a **deeper plexus**. Often **ligated at radical prostatectomy** to limit bleeding, but **ligation can damage the sphincter, particularly laterally**.

- **Santorini's (prostatic venous) plexus:** drains the prostate and part of the rectum; met on entering the **lateral prostatic fascia** at radical prostatectomy; drains, with several inferior vesical veins, to the **internal iliac vein**.

Pelvic Lymphatics

- Pelvic viscera drain to the iliac nodes and their tributaries, with **frequent cross-drainage between right and left**.
- **External iliac nodes:** from the top of the external iliac artery distally to just over the pelvic brim; drain the **anterior abdominal wall, urachus and bladder**.
- **Node of Cloquet:** the most distal node of the packet, where the external iliac vessels leave the pelvis beneath the inguinal ligament (in the femoral canal); the **distal limit of PLND**.
- **Internal iliac nodes:** drain **prostate and bladder**; the **presacral and obturator nodes** drain into the internal iliac chain.
- **External genitalia → inguinal nodes → iliac nodes.**

Pelvic Innervation

- Sensation to pelvis and genitalia comes from the **lumbosacral plexus**.
- **Iliohypogastric (L1):** anterior abdominal wall. **Ilioinguinal (L1):** through the inguinal canal → **anterior scrotal skin**.
- **Genitofemoral (L1–L2):** on the anterior surface of psoas — the **lateral landmark/border of PLND**; genital branch through the inguinal canal → cremaster and anterior scrotum; femoral branch → anterior thigh (Figure 102.14).

TABLE 102.2 SOMATIC AND AUTONOMIC NERVES OF THE PELVIS (REBUILT)

Nerve	Roots	Supplies	Anatomical course
Iliohypogastric	L1	Anterior abdominal wall and pubis	Between internal oblique and transversus
Ilioinguinal	L1	Anterior scrotum	Through inguinal canal
Genitofemoral	L1, L2	Genital branch: cremaster, anterior scrotum; femoral branch: anterior thigh	On anterior psoas; genital branch through inguinal canal; femoral branch with external iliac under inguinal ligament
Lateral cutaneous femoral	L2, L3	Lateral thigh	Lateral to psoas, on iliacus
Posterior femoral cutaneous	S2, S3	Perineum and posterior scrotum	Greater sciatic foramen
Pudendal	S2–S4	Sensory: perineum, scrotum, penis; motor: levator ani, urinary and anal sphincters	Follows internal pudendal artery
Pelvic somatic efferents	S2–S4	Motor to levator ani and striated urethral sphincter	Branches of pudendal nerve into perineum
Nervi erigentes	S2–S4	Parasympathetic fibres to pelvic viscera	Sacral cord → bladder and rectum

- **What the table is for:** root values and territories. **Pattern:** **L1–L2 nerves serve the groin and scrotum anteriorly** (and run through or past the inguinal canal); **S2–S4 serve the perineum, sphincters and viscera**. **Trap:** anterior scrotum is L1 (ilioinguinal, genitofemoral), posterior scrotum is sacral (pudendal, posterior femoral cutaneous).

Motor Nerves

FEMORAL VERSUS OBTURATOR NERVE

	Femoral nerve	Obturator nerve
Roots	L2–L4	L2–L4
Function	Sensation to anterior thigh; motor to thigh extensors	Thigh adduction
Course	Deep within psoas	First seen between internal iliac vessels and pelvic sidewall at the common iliac bifurcation; runs anteriorly through the obturator canal
Injury	Psoas hitch; retractor-blade pressure at laparotomy; inguinal node dissection	PLND → difficulty with thigh adduction

- **Sacral plexus:** lumbosacral trunk (**L4, L5**) + sacral segmental nerves; lies between the internal iliac vessels and piriformis; leaves through the **greater sciatic foramen** to supply the lower leg and posterior thigh.

Autonomic Nerves of the Pelvis

SYMPATHETIC VERSUS PARASYMPATHETIC PELVIC OUTFLOW

	Sympathetic	Parasympathetic
Origin	Lateral horn, T10–L2	Lateral horns, S2–S4 (pelvic splanchnic nerves, nervi erigentes)
Route	Superior hypogastric plexus (continuation of coeliac plexus and lumbar sympathetic fibres) → two hypogastric nerves , anterior to the aorta and medial to the internal iliac artery, entering the pelvis beneath the endopelvic fascia; plus the sympathetic trunks , lateral to the vertebral bodies and behind the iliac vessels, terminating in front of the coccyx	Join the hypogastric nerves to form the pelvic (inferior hypogastric) plexus

- **Pelvic plexus:** on either side of the rectum, **centred on the seminal vesicles**; supplies rectum, bladder neck and distal ureter (Figure 102.16).
- Distally it supplies the prostate and continues as the **cavernosal nerves**, which run **posterolateral to the prostate on the surface of the rectum, outside Denonvilliers fascia**. With the prostatic arteries and veins they form the **neurovascular bundle (NVB)**.
- At the prostatic apex the NVBs lie posterolaterally at about the **5 and 7 o'clock positions**, then branch to the **striated sphincter and corpora cavernosa** (erectile tissue).
- **Damage to the NVB or pelvic plexus at prostatectomy or rectal resection → erectile dysfunction.**
- *Figures 102.15–102.16* — sympathetics descend from above via the superior hypogastric plexus and hypogastric nerves while pelvic parasympathetics ascend from S2–S4; they meet in the inferior hypogastric plexus, which in lateral view sits beside the seminal vesicle with superior and inferior vesical pedicles and the NVB running to the prostate (Schlegel and Walsh, 1987).

KEY POINTS: PELVIC CIRCULATION AND INNERVATION

- The pelvic vascular supply derives from the common iliac vessels and their branches; the deep pelvis is supplied by the internal iliac vessels.
- There is frequent cross-drainage between lymphatics of both sides of the pelvis.

- Prostatic arteries, veins and the cavernosal nerves form the neurovascular bundle, which supplies erectile tissue.

Concept — *The deep pelvis hangs off the internal iliac artery and the pelvic plexus, both carried in the endopelvic fascia beside the viscera. PLND hazards follow the external iliac–obturator axis (vein confluence, accessory obturator vessels, obturator nerve), whereas erectile function depends on one posterolateral bundle formed where plexus meets prostatic vessels.*

PELVIC VISCERA

Rectum

- Begins at **S3**, follows the sacral curve through the pelvic floor and ends in the perineum as the anus.
- Anterior rectum is peritoneum-covered up to the **rectovesical pouch**; incising this peritoneum gives access to the **seminal vesicles behind the bladder** — the **posterior, pelvic fascia-sparing approach to robotic-assisted radical prostatectomy** (Figure 102.17).
- **Beyond the pouch, Denonvilliers fascia separates rectum from prostate.**
- **Blood supply:** middle rectal artery (internal iliac) and inferior rectal artery (internal pudendal). The **pelvic plexus lies laterally** and provides autonomic innervation.
- **Distal rectum (ampulla)** — site of the **rectourethralis** muscle; lies near the striated urethral sphincter and prostate and is the **commonest site of rectal injury at prostatectomy.**

Bladder

- Covered by peritoneum superiorly; **mostly extraperitoneal**. The **dome is continuous with the urachus** — a fibrous embryological structure, typically epithelium-lined but obliterated, running in the midline to the umbilicus.
- The urachus can be a source of **bladder adenocarcinoma**; persistent communication with the urinary tract → **fistulae or cysts.**
- **Anterior — the retropubic space:** a potential extraperitoneal space deep to transversalis fascia, often developed surgically to reach bladder or prostate. **Lateral** — loose connective tissue and fat. **Posterior** — ureters entering alongside the seminal vesicles and vasa (Figure 102.18); the rectum lies behind these.
- **Adult bladder rests in the true pelvis;** its neck is firmly attached to the prostatic base, and its urothelium is continuous with the prostatic urethra. **In infants the bladder is intra-abdominal** and migrates into the true pelvis during puberty.
- **Wall layers:** urothelium (relatively impermeable, multilayered transitional epithelium with superficial **umbrella cells**) → lamina propria with **muscularis mucosae** → **muscularis propria:** smooth muscle in three layers, **inner longitudinal, middle circular, outer longitudinal**, forming a network for coordinated emptying.
- At the **bladder neck** the longitudinal fibres are continuous with the prostatic urethra; the **circular fibres form the internal urethral sphincter**, an important component of male continence — seen in **post-prostatectomy incontinence** when they are damaged.
- The bladder neck is **adrenergically innervated**; closure enables **antegrade ejaculation.**
- The **trigone is the thickest part of the detrusor**; outer longitudinal fibres give a strong muscular backing; anteriorly the longitudinal fibres are continuous with the **puboprostatic ligaments.**

Ureterovesical Junction

- The ureter is encased in fibrous tissue — **Waldeyer's sheath** — as it enters the trigone (Figure 102.19). Entry is **cephalad and lateral**; the ureter runs **1.5–2 cm** to an orifice lying **caudal and medial** to its point of insertion.
- The **intramural ureter is narrow** → a site of obstruction by stones or by marked bladder wall thickening.
- **Flap-valve**: strong detrusor backing **occludes the intramural ureter during filling** → one-way flow. **Vesicoureteral reflux (VUR)** is thought to arise partly from **weak or shortened muscular backing**.
- The ureteric entry point is a **weak spot in the detrusor** → **Hutch diverticulum** under raised intravesical pressure.

SUPERFICIAL VERSUS DEEP TRIGONE

	Superficial trigone	Deep trigone
Derived from	Longitudinal smooth muscle of the ureters , joining in a triangle	Waldeyer's sheath continued into the bladder + detrusor
Anchored at	Verumontanum	Bladder neck
Role	Superficial layer of trigone	Outer layers; anchor ureters to the bladder → compression of intramural ureter during filling → prevents VUR

Figure 102.19 after Tanagho and Pugh, 1963: in section, the ureter passes obliquely through the detrusor with a submucosal segment and detrusor backing; Waldeyer's sheath surrounds the prevesical ureter and extends inward as the deep trigone.

Blood Supply and Lymphatic Drainage of the Bladder

- Besides the vesical branches, several other hypogastric branches contribute to the **bladder pedicles**, posterior and lateral to the bladder. The veins form a **plexus within the pedicles** draining to the **internal iliac vein**.
- Some nodes lie in the perivesical space, but bladder lymphatics **begin in the lamina propria** and drain **mainly to the external iliac nodes**, also to internal iliac and obturator nodes. **Significant cross-drainage to both sides** of the pelvis.

Innervation of the Bladder

- Efferents come from the **anterior portion of the pelvic plexus**.
- **High density of parasympathetic, cholinergic endings**; **relatively little sympathetic** innervation of the body.
- **Nonadrenergic noncholinergic (NANC) fibres** — probably **purinergic**; small contribution to normal contraction, but potential **pharmacological targets in pathological states**.
- **Bladder neck**: dense **$\alpha 1$ -adrenergic receptors in males** → closure for **antegrade ejaculation** and continence. **Nitric oxide synthase-containing neurons** in bladder neck and trigone may promote **relaxation during micturition**.
- **Afferents** travel with the hypogastric plexus (and pelvic nerves) to the dorsal root ganglia.

Pelvic Ureter

- Detailed ureteric structure is covered in Chapter 88. The pelvic ureter **begins where it crosses the common iliac artery**; proximity to the iliac vessels gives a **relative narrowing that can obstruct stone passage**.
- **Extrinsic obstruction**: gravid uterus; pathologically enlarged iliac lymph nodes.

- **Course:** crosses iliac vessels → laterally towards the sidewall → **beneath the umbilical artery** → turns medially **behind the vas deferens** → bladder.
- Retroperitoneal, but **peristalsis may be visible through the peritoneum** in thin patients, especially at the iliac vessels.
- **Blood supply:** common iliac artery plus several internal iliac branches (Table 102.3), **entering laterally** → **preserve the lateral attachments** to avoid devascularisation. Adventitial (intramural) vessels permit circumferential dissection if needed, but the supply is variable → **stricture after circumferential dissection or ligation**.
- **Lymphatics:** common, external iliac and internal iliac nodes.

KEY POINTS: PELVIC VISCERA

- The rectum is separated from the bladder by a fascial layer (Denonvilliers fascia).
- The bladder is mostly extraperitoneal except for the dome.
- The anterior retropubic space is an extraperitoneal access point to bladder and prostate.
- The ureters are anchored to the strong muscular backing of the trigone, which prevents reflux of urine.

Concept — *Bladder outlet competence and ureteric anti-reflux are both muscular-backing mechanisms: circular fibres under $\alpha 1$ control shut the neck (continence, antegrade ejaculation), and trigonal/detrusor backing flattens the intramural ureter (anti-reflux). Lose the backing and the result is, respectively, incontinence or retrograde ejaculation, and reflux or a Hutch diverticulum.*

ENDOSCOPIC ANATOMY

- The male bladder and urethra are readily examined in **awake patients under local analgesia**. **Flexible fiberoptic cystoscopy** visualises the whole bladder, including ureteric orifices and trigone → identifies sources of **haematuria** and localises **bladder tumours**.
- **Sequence:** survey of bladder → **retroflexion at the bladder neck** (intravesical prostatic protrusion; small tumours hidden just inside the neck) → prostatic urethra: **verumontanum posteriorly**, ejaculatory ducts on its summit → withdraw distally → **coaptation of the external sphincter just beyond (distal to) the verumontanum** → bulbar urethra (Figure 102.20).
- The **verumontanum is the key landmark at transurethral resection of the prostate (TURP)** for benign prostatic enlargement: **deep resection beyond it can damage the sphincter → incontinence**.

Concept — *The verumontanum is the sentinel for the external sphincter: everything proximal belongs to the prostate and may be resected; the sphincter begins immediately distal to it.*

RADIOGRAPHIC IMAGING OF THE PELVIS

Pelvic Radiographs and Fluoroscopy

- Bony landmarks are shown in high detail, but **pelvic organs are poorly depicted** because of similar densities. A partly filled bladder may appear as a faint line and rounded density above the symphysis; rectum and sigmoid are inferred from luminal gas or stool.
- Calcifications are well seen. A **phlebolith** (calcification in a pelvic vein lumen) is **essentially a normal, common finding** with no clinical relevance — but it **mimics a distal ureteric or bladder stone**.
- **Cystogram:** the empty bladder lies at or just above the symphysis; on filling it rises into the false pelvis and spreads laterally and anteriorly towards the sidewalls (Figure 102.2B). **Absent lateral distension → a**

space-occupying mass or collection, classically an **extraperitoneal pelvic haematoma in pelvic trauma**, or a **lymphocele or haematoma after radical prostatectomy**.

- On voiding, the **posterior urethra is a well-distended tube** because both involuntary and voluntary sphincters relax completely — so the **posterior urethra is best imaged by voiding cystourethrography (VCUG)**.
- **Retrograde urethrography (RUG)** is performed optimally in the **oblique position**; the membranous and prostatic urethra can be hard to opacify on RUG because of voluntary external sphincter contraction, but **steady injection pressure** usually achieves it.
- **Posterior urethra = membranous + prostatic**. The **membranous urethra** crosses the urogenital diaphragm; it is the **shortest and narrowest segment**, just proximal to the cone of the bulbous urethra, and **must not be mistaken for a stricture**. The **prostatic urethra** (bladder base → urogenital diaphragm) is tubular or somewhat flattened, identified by the **verumontanum as an oval central filling defect**.
- Once the prostatic urethra fills, contrast should flow freely into the bladder — **triangular** in a decompressed bladder, a vague diluted pattern in a full one.
- **Glands of Littre and Cowper's ducts** occasionally opacify beside the urethra — **not injuries or fistulae**.

VOIDING CYSTOURETHROGRAPHY VERSUS RETROGRADE URETHROGRAPHY

	VCUG	RUG
Sphincters	Relaxed during voiding	Voluntary external sphincter may contract against injection
Best for	Posterior urethra — well-distended tube	Urethra distal to the sphincter; posterior urethra opacified only with steady pressure
Technique point	Oblique spot images during voiding (Figure 102.2B)	Oblique position

Computed Tomography

- Computed tomography (CT) is acquired as a **volumetric data set**, reconstructable in any plane (usually axial, coronal, sagittal).
- **High spatial resolution** but **relatively low contrast resolution** compared with magnetic resonance imaging (MRI), because soft tissues attenuate similarly → most CT needs **intravenous contrast**. **Exception — urolithiasis**: stones attenuate strongly, so **no intravenous contrast is needed**.
- **Male pelvic CT uses**: pain, infection, stones, postoperative complications, **staging of known malignancy**. **Poor for detecting primary urological tumours** — most seen only when locally advanced.
- **Spaces**: peritoneum reflects over the bladder dome; much of the bladder is surrounded by extraperitoneal space. The **retropubic space** (anterior) communicates with the **lateral perivesical spaces** (fat and vessels) → where urine collects in **extraperitoneal bladder rupture** and where advanced bladder cancer infiltrates.
- From the dome the peritoneum continues into the **retrovesical space** as a pelvic reflection of variable depth between bladder and rectum; being the **most dependent part of the peritoneum**, it is a common site of **fluid collection and postoperative abscess**.

LYMPH NODES ON CT: NORMAL VERSUS MALIGNANT

	Normal	Malignant
Shape / outline	Smooth, oval or elongated	Rounded ; irregular or lobulated borders with extranodal extension
Internal	Homogeneous soft tissue; fatty hilum where vessels enter	Heterogeneous or hypervascular

	Normal	Malignant
Size	Commonest criterion; highest specificity at ≥ 1 cm short axis	
	• Nodes are well seen because of CT's spatial resolution and lie in named groups adjacent to the major pelvic arterial divisions (Figure 102.21). • Multiphasic CT: iodinated contrast is excreted almost exclusively by the kidneys, so delayed (excretory) images at 7–10 minutes fill ureters and bladder → CT urogram evaluates the entire collecting system, ureteric course and wall, bladder wall and contents in one examination. • Ureter on CT urogram (Figure 102.22): enters the pelvis crossing anteriorly and obliquely over the common or external iliac artery near the bifurcation → physiological narrowing from vascular compression and the turn over the brim; undulation from peristalsis; unopacified segments from peristalsis — not disease. It then runs laterally along the sidewall and makes an abrupt medial and anterior turn into the ureterovesical junction. • Non-contrast thin-slice CT: the ureter can almost always be followed from pelviureteric junction to bladder; pelvic veins may mimic it but branch, disappear or take the wrong course on following slices.	

Magnetic Resonance Imaging

- Resolution is similar to CT, but MRI's **higher soft tissue contrast resolution** gives more detailed muscle and soft tissue anatomy: organs, pelvic floor muscles, fascia, vessels and nerves.
- Multiplanar; combines T1- and T2-weighted images with and without **fat saturation**. Interpret **as a set** across T1, T2 and postcontrast sequences.

T1- VERSUS T2-WEIGHTED MRI IN THE PELVIS

T1-weighted	T2-weighted
Bone and muscle; postcontrast enhancement (gadolinium affects T1 far more than T2)	Organ anatomy and muscle; very fluid-sensitive → oedema, inflammation, collections; best anatomical assessment of prostate, seminal vesicles and supports
• Bladder: urine T2-hyperintense, wall T2-dark. Wall thickness varies with distension; a normal well-distended bladder has a wall <3 mm; >3 mm is nonspecific pathological thickening. • Prostatic urethra: followed through the gland on axial and sagittal images, usually in the midline posterior third (Figure 102.23). The verumontanum appears in the mid-prostatic urethra on axial images as a focal triangular or rounded bulge or filling defect (Figure 102.24) — where the ejaculatory ducts open, though the ducts (and the prostatic ducts) are usually not seen. • External urethral sphincter: just below the prostatic apex, within the urogenital diaphragm — a T2-dark doughnut of variable thickness (Figure 102.25). Cowper's glands are not seen unless enlarged or cystic. • Prostate MRI is covered in Chapters 6, 155 and 157. • Pelvic floor = levator ani complex (puborectalis, pubococcygeus, iliococcygeus). Puborectalis: U-shaped, best on axial images just above the external anal sphincter, from the inferoposterior pubis around the anorectal junction; important for voluntary urinary and faecal continence; function assessable on dynamic pelvic floor MRI. Pubococcygeus and iliococcygeus: thin broad sheets to the coccyx, best on coronal images; fibres and organ insertions poorly shown. • Endopelvic fascia appears as fat signal on T1 or T2, with faint connective tissue strands. • Anal sphincters (below the floor, posteriorly): internal — thick doughnut, more T2-hyperintense; external — thin, more vertically oriented, darker (Figure 102.26). • Penile base (anteriorly, below the urogenital diaphragm): corpora cavernosa T2-bright, almost fluid intensity (vascular content); corpus spongiosum similar, with the bulbar urethral wall as a faint T2-dark	

outline; **ischiocavernosus and bulbospongiosus** as thin, linear T2-dark fibres beside the cavernosum and spongiosum respectively.

IMAGING MODALITIES FOR THE MALE PELVIS

Modality	Strength	Weakness / normal mimics
Plain radiograph	Bony pelvis ; calcifications	Organs poorly shown; phlebolith mimics stone
Fluoroscopy (cystogram, VCUG, RUG)	Urethra and bladder contour, dynamically	Membranous narrowing mimics stricture; glands of Littre and Cowper's ducts mimic injury
CT	Spatial resolution : stones (no contrast), lymph nodes, collecting system (CT urogram)	Low contrast resolution; poor for primary tumours; peristaltic gaps mimic disease
MRI	Contrast resolution : prostate, seminal vesicles, pelvic floor (T2)	Small muscle insertions and ducts not seen

KEY POINTS: RADIOGRAPHIC IMAGING OF THE PELVIS

- Radiographs provide high-resolution images of the bony pelvis.
- The urethra is best imaged by fluoroscopy.
- CT is better for urolithiasis, lymph nodes and the urinary system.
- MRI has better soft tissue resolution for the prostate and pelvic floor.

Concept — *Modality choice follows the **spatial-versus-contrast-resolution trade-off**, and almost every "normal mimic" in this section — phlebolith, membranous narrowing, peristaltic gaps, opacified glands — is a trap for overcalling disease in a normal pelvis.*